

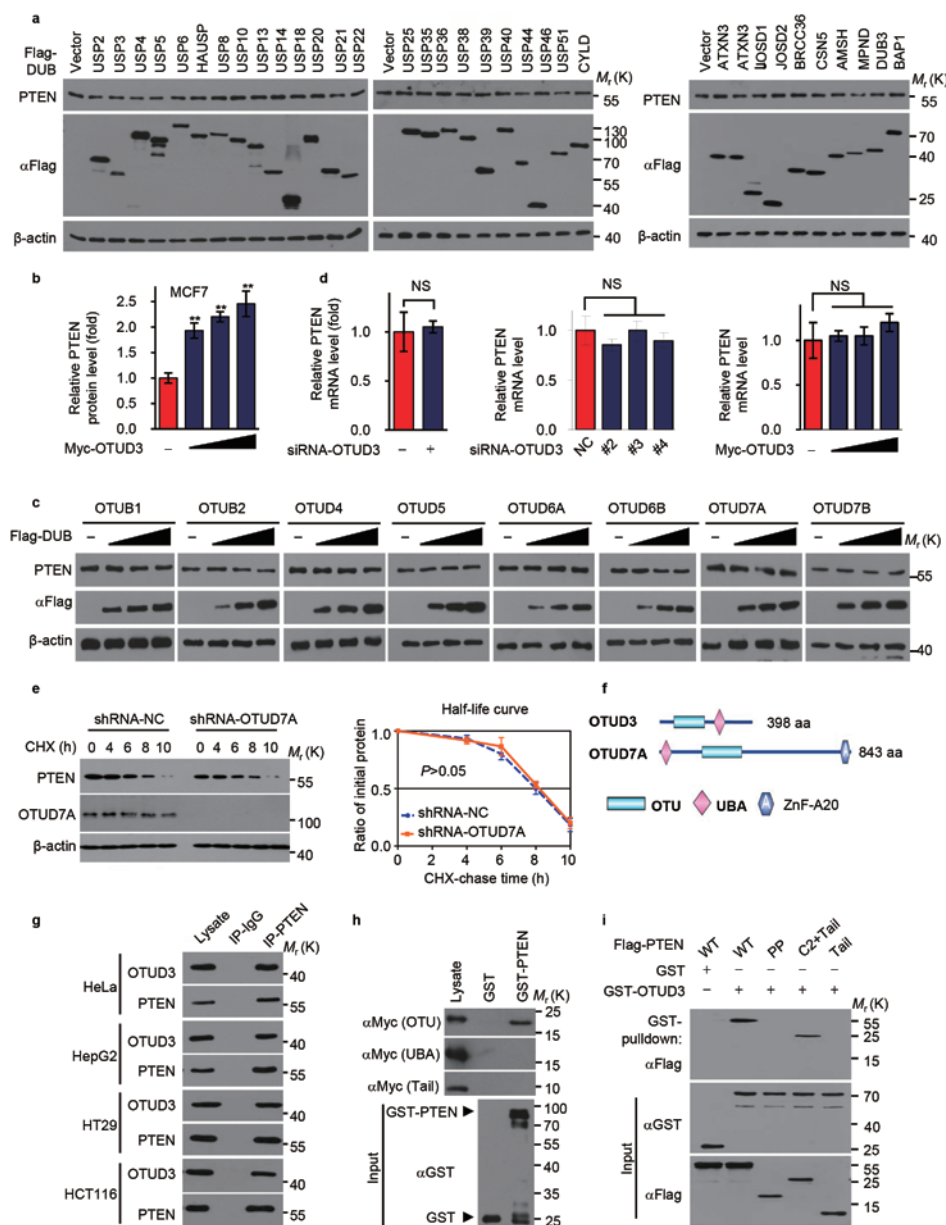
Correction notice

Nature Cell Biology <http://dx.doi.org/10.1038/ncb3218>
(2015)

Deubiquitylase OTUD3 regulates PTEN stability and suppresses tumorigenesis

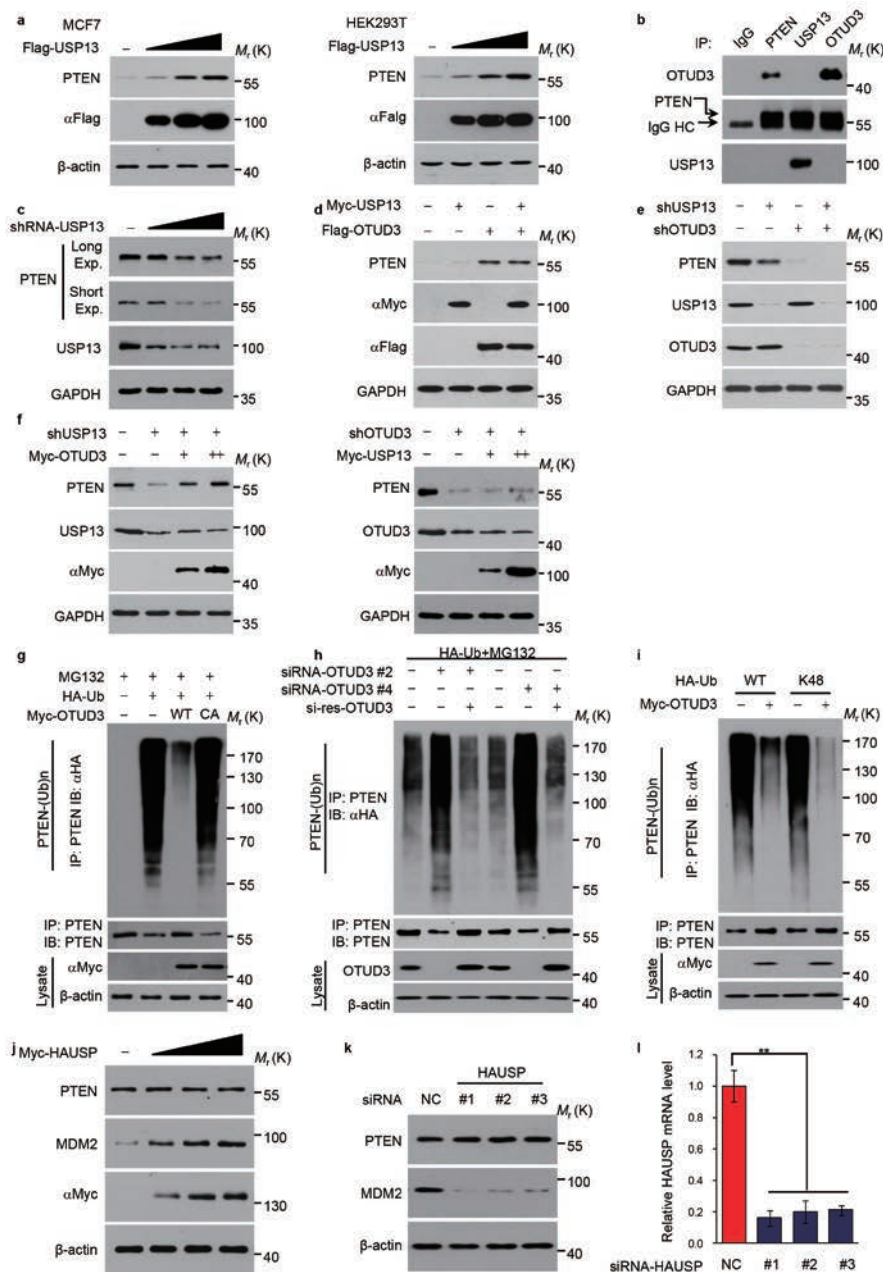
Lin Yuan, Yanrong Lv, Hongchang Li, Haidong Gao, Shanshan Song, Yuan Zhang, Guichun Xing, Xiangzhen Kong, Lijing Wang, Yang Li, Tao Zhou, Daming Gao, Zhi-Xiong Xiao, Yuxin Yin, Wenyi Wei, Fuchu He and Lingqiang Zhang

In the original version of this file the figures appeared in the wrong order. This error was corrected on 18 August 2015.



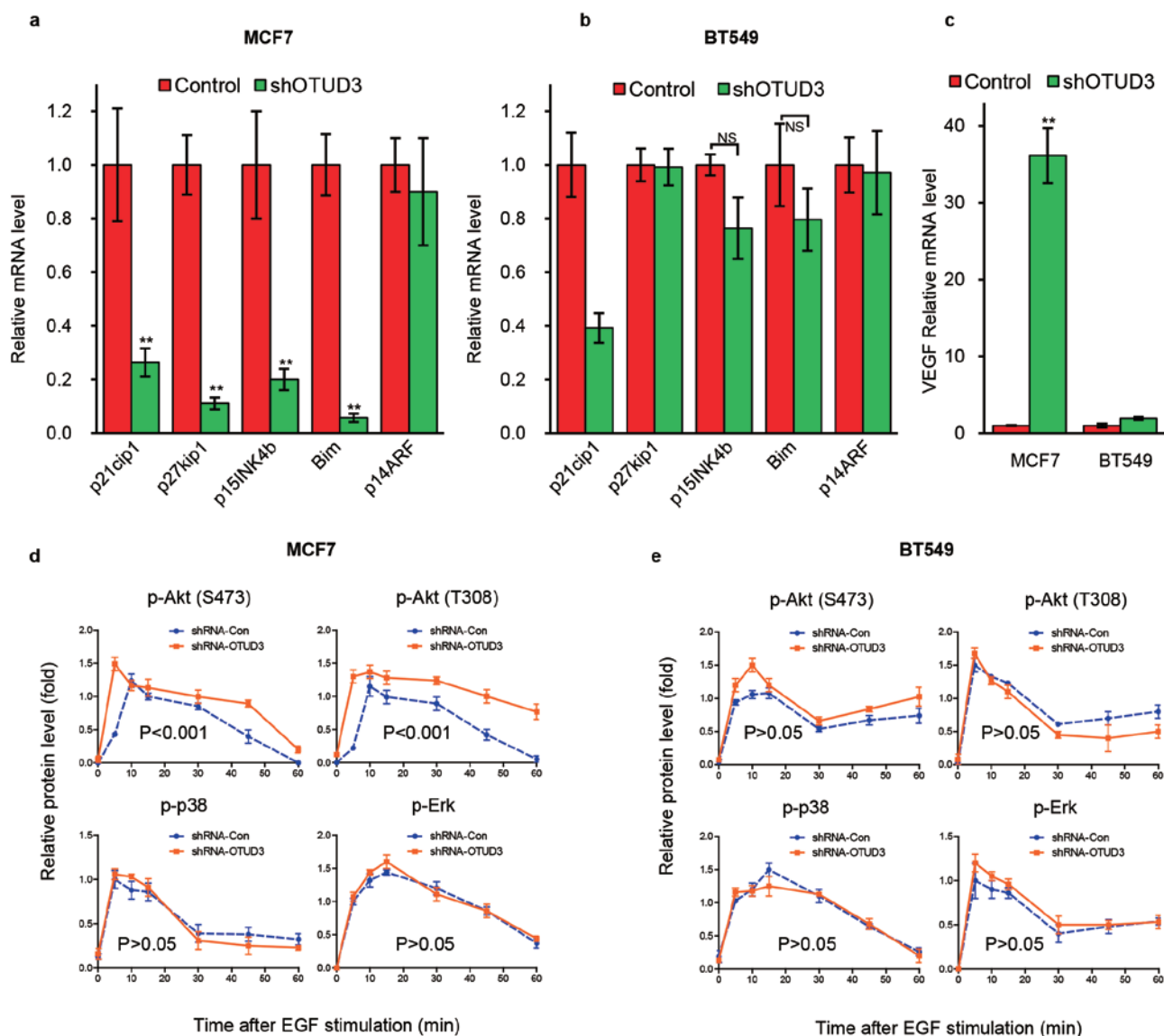
Supplementary Figure 1 OTUD3 stabilizes PTEN protein. **a**, Protein level analysis of PTEN in the presence or absence of overexpressed DUBs. HCT116 colon cancer cells were overexpressed with each of the Flag-tagged DUBs. **b**, Quantification of the PTEN protein levels relative to β-actin for Fig. 1b. $n=3$ independent experiments (**: $p < 0.01$, Student's t -test). **c**, Increasing amounts of plasmids encoding other members of OTU sub-family of DUBs were each transfected into HCT116 cells. The endogenous PTEN protein levels were examined by western blot. **d**, Knockdown of OTUD3 by pool siRNAs (left) or individual siRNAs (middle) or overexpression of OTUD3 (right) had no significant effect on PTEN mRNA level in MCF7 cells. $n=3$ independent experiments (NS: no significance, Student's t -test). **e**, MCF7 cells stably expressing the indicated shRNA were treated with cycloheximide (10 $\mu\text{g ml}^{-1}$), and harvested at the indicated times. The left panels show

immunoblots of PTEN and OTUD7A. Right panel: quantification of the PTEN protein levels relative to β-actin. $n=3$ independent experiments (NS: no significance, two-way ANOVA test). **f**, Domain structure of OTUD3 and OTUD7A. **g**, Co-immunoprecipitation assays in four types of cancer cell lines to detect the endogenous interactions. The total cell lysates were immunoprecipitated with anti-OTUD3, anti-PTEN or control IgG as indicated. Both the lysates and the immunoprecipitates were subjected to western blot analysis. **h**, GST pull-down assays were performed to map the domain of OTUD3 required for interaction with PTEN. **i**, GST pull-down assays were performed to map the domain of PTEN required for interaction with OTUD3. All panels are representative results of three or more independent experiments. Statistics source data can be found in Supplementary Table 4. Uncropped images of blots are shown in Supplementary Fig. 9.



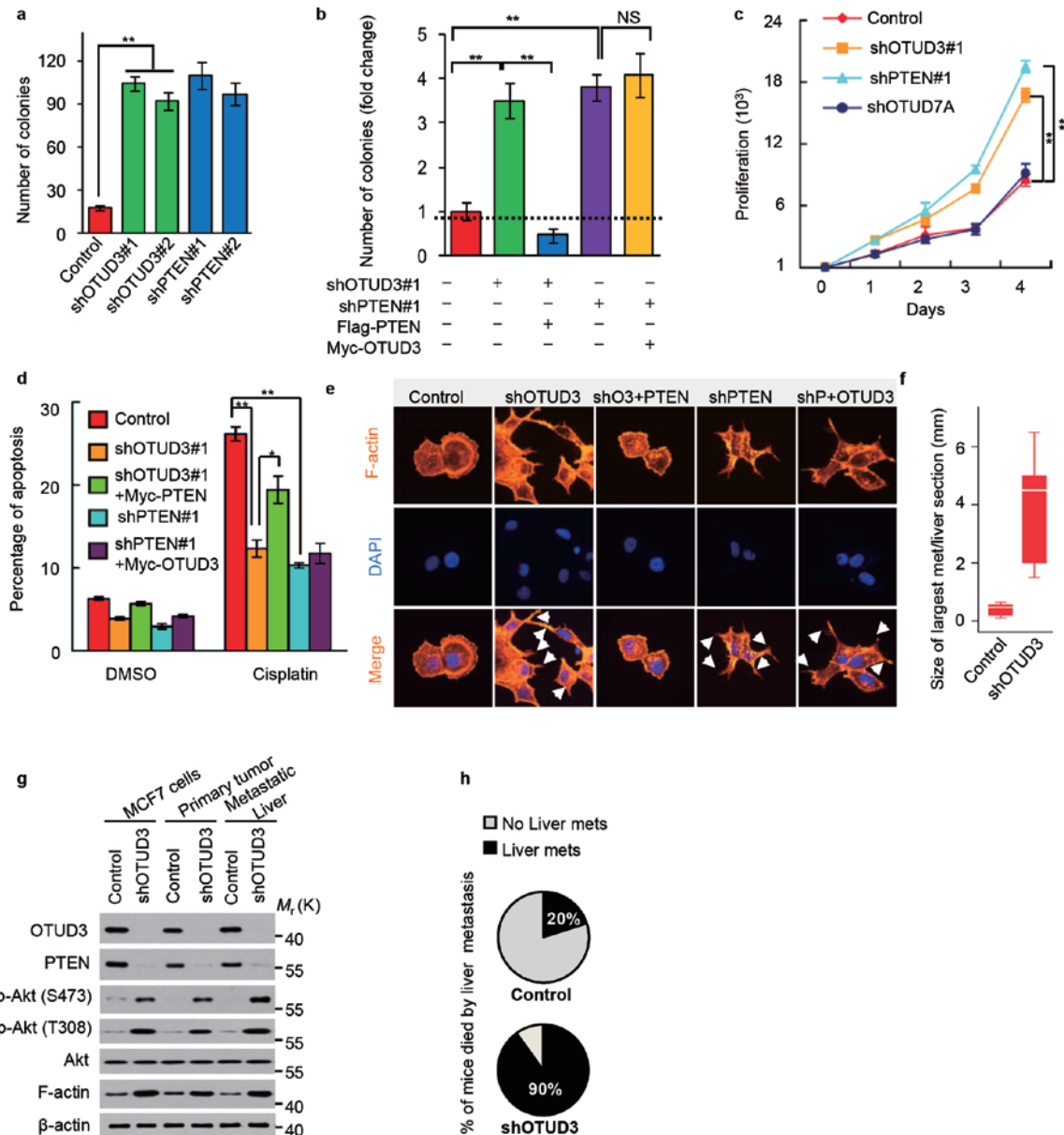
Supplementary Figure 2 OTUD3 is a more potent regulator of PTEN than USP13, and HAUSP has no significant effect on PTEN protein level. **a**, Increasing amounts of Flag-USP13 were transfected into MCF7 (left) and HEK293T (right) cells. After 48 h, PTEN protein levels were detected. **b**, Co-immunoprecipitation of endogenous PTEN, USP13 and OTUD3 in MCF7 cells. Protein extracts were immunoprecipitated with antibody against PTEN, USP13 or OTUD3, followed by immunoblotting with antibodies indicated. **c**, Increasing amounts of shRNA-USP13 were transfected into MCF7 cells. After 72 h, PTEN protein levels were detected. **d**, Protein level analysis of PTEN in the presence of overexpressed USP13 or OTUD3. MCF7 cells were overexpressed with Myc-USP13 or Flag-OTUD3. **e**, Protein level analysis of PTEN in MCF7 cells stably expressing the indicated shRNA. **f**, MCF7 cells were transfected with a constant amount of shRNA-USP13 and increasing amounts of Myc-OTUD3 (left), or a constant amount of shRNA-OTUD3 and increasing amounts of Myc-USP13 (right). After 48 h, PTEN protein levels were detected. **g**, HCT116 cells transfected with

the indicated constructs were treated with MG132 for 8 h before harvest. PTEN was immunoprecipitated with anti-PTEN and immunoblotted with anti-HA. **h**, siRNA-resistant OTUD3 was introduced into the HCT116 cells together with the OTUD3 siRNA. PTEN ubiquitination was measured. **i**, The PTEN ubiquitination linkage was analyzed in HCT116 cells transfected with OTUD3 and the indicated ubiquitin WT or K48-only plasmids. **j**, Increasing amounts of HAUSP plasmids were transfected into the MCF7 cells and the lysates were subjected to Western blot analysis. **k**, Knockdown of HAUSP by individual siRNAs had no significant effect on PTEN protein level but downregulated MDM2 protein level (positive) in MCF7 cells. **l**, Effectiveness of HAUSP knockdown by individual siRNAs was measured by quantitative RT-PCR. Data are shown as mean $\pm$ s.d. of $n=3$ independent experiments. (**: $p < 0.01$, Student's t -test). All panels are representative results of three or more independent experiments. Statistics source data can be found in Supplementary Table 4. Uncropped images of blots are shown in Supplementary Fig. 9.



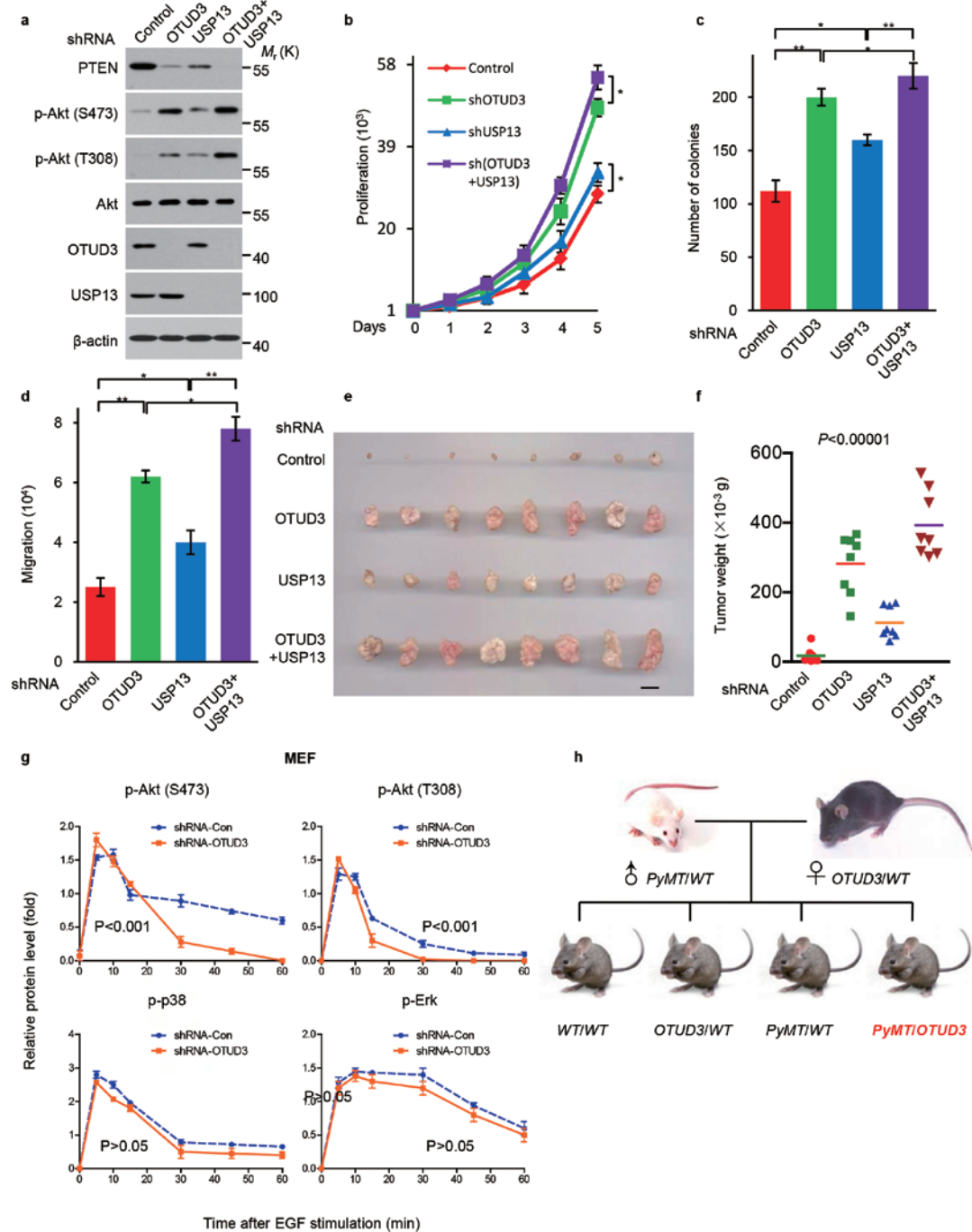
Supplementary Figure 3 OTUD3 negatively regulates Akt signaling in a PTEN-dependent manner. a-b, Quantitative RT-PCR analysis of indicated gene mRNA level in MCF7 (PTEN wild-type) (a) or BT549 (PTEN-null) (b) cells stably expressing the shRNA-Control or shOTUD3#1. Data were analyzed using two-tailed unpaired Student's *t*-test. Data are shown as mean $\pm$ s.d., *n*=3 independent experiments. *: *p* < 0.05, **: *p* < 0.01, NS: no significance. c, Quantitative RT-PCR analysis of VEGF gene mRNA level in MCF7 and BT549 cells stably expressing shRNA-Control or shOTUD3 #1.

Data were analyzed using two-tailed unpaired Student's *t*-test. Data are shown as mean $\pm$ s.d., *n*=3 independent experiments. d-e, OTUD3 shRNA-transduced MCF7 or BT549 cells were serum-starved and treated with EGF for various times. Quantification of the phosphorylation levels of Akt, p38 and Erk protein relative to total Akt, p38 and Erk protein levels for Fig. 3f and 3g. Data are shown as mean $\pm$ s.d., *n*=3 independent experiments (two-way ANOVA test). Statistics source data can be found in Supplementary Table 4.



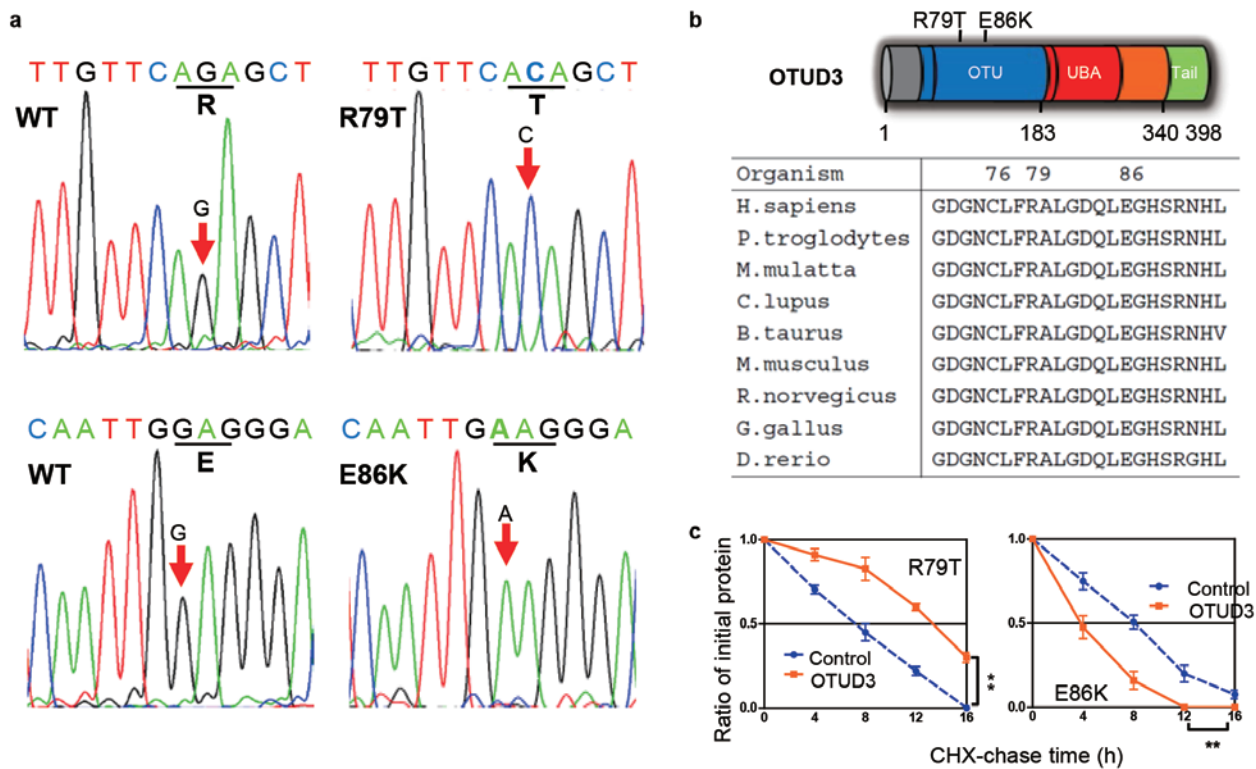
Supplementary Figure 4 OTUD3 inhibits cell proliferation, promotes cell apoptosis, and suppresses cell migration. **a**, Anchorage-independent growth of MCF7 cells stably expressing the indicated shRNA on soft agar. Viable colonies after 3 weeks were counted and the data from $n=3$ independent experiments were presented (mean $\pm$ s.d.). **: $p < 0.01$, Student's t -test. **b**, MCF7 cells stably expressing control shRNA, OTUD3 shRNA or PTEN shRNA were transfected with either OTUD3 or PTEN as indicated. Cells were tested for growth in colony assay. Colonies after one week were counted and the data (mean $\pm$ s.d.) from $n=3$ independent experiments were presented (**: $p < 0.01$, NS: no significance, Student's t -test, compared with cells expressing control shRNA;). **c**, MCF7 cells stably expressing control shRNA, OTUD3 shRNA or PTEN shRNA were seeded and cell proliferation was measured by MTS assay for 5 days. Data are shown as mean $\pm$ s.d.. $n=3$ independent experiments (**: $p < 0.01$, two-way ANOVA test). **d**, The indicated HCT116 cells were treated with cisplatin or DMSO to induce

apoptosis. The percentage of apoptotic cells was measured by Annexin-V staining. Data are shown as mean $\pm$ s.d.. $n=3$ independent experiments (**: $p < 0.05$; *: $p < 0.01$, Student's t -test). **e**, The cellular morphology was analyzed in MCF7 cells stably expressing the indicated shRNA and/or transfected with the OTUD3 or PTEN plasmids. Cells were fixed and stained for F-actin (red). Nuclei were stained with DAPI (blue). Scale bar, 20 μ m. **f**, Box plots showed the distribution of the size (largest in each section) of metastasis per section. Data were analyzed using Mann-Whitney test ($n=50$ fields from 10 independent mice, $p=0.0025$). **g**, Immunoblotting of OTUD3, PTEN, p-Akt, Akt, F-actin and β -actin in lysates of primary tumors and metastatic livers from mice injected with the stable shRNA-control or shRNA-OTUD3 MCF7 cells. **h**, Pie charts for mice died by liver metastasis. Data are representative results of 3 independent experiments. 3×10^3 Statistics source data can be found in Supplementary Table 4. Uncropped images of blots are shown in Supplementary Fig. 9.



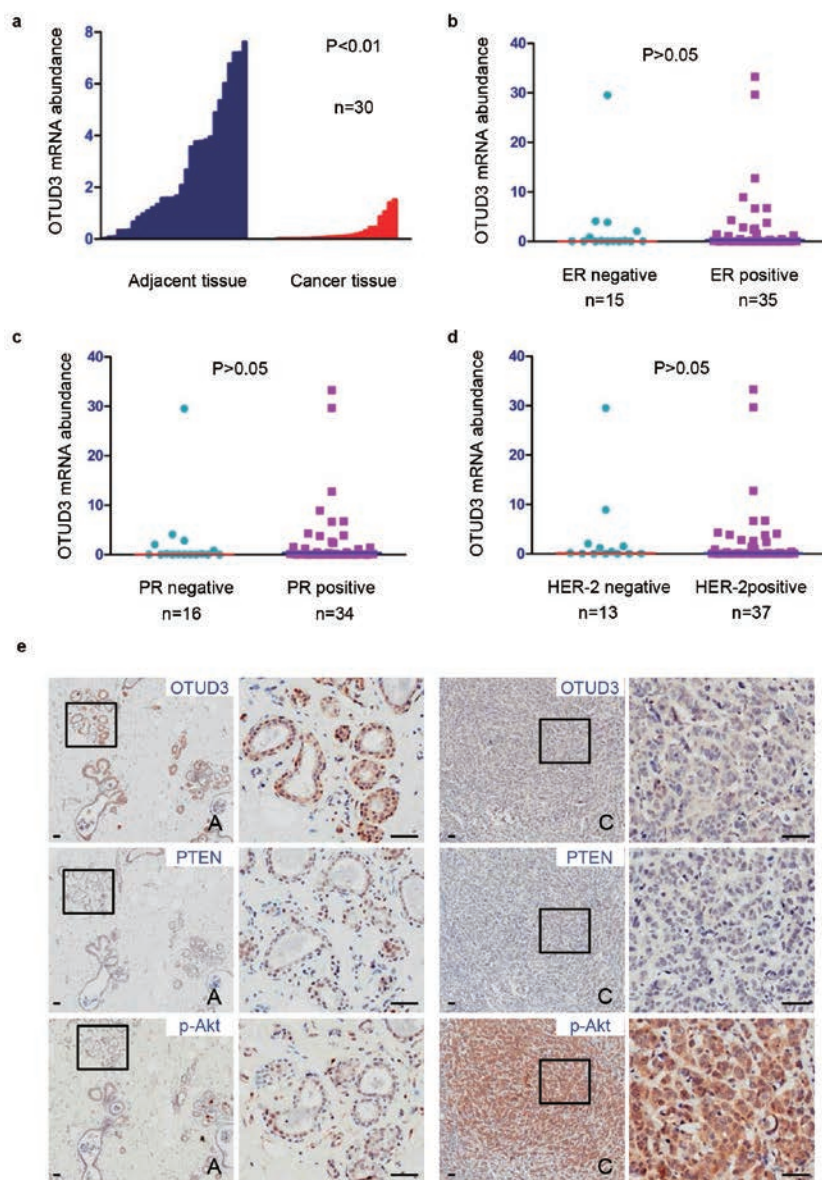
Supplementary Figure 5 The possible synergy between OTUD3 and USP13, and the phenotype of OTUD3 transgenic mice. **a**, Akt phosphorylation was analyzed in MCF7 cells stably expressing the indicated shRNA. **b–d**, The MCF7 cells expressing OTUD3 WT or mutants were measured for cell proliferation (**b**, two-way ANOVA test), anchorage-independent growth in soft agar (**c**, Student's *t*-test) and transwell migration assay (**d**, Student's *t*-test). Data are shown as mean $\pm$ s.d.. $n=3$ independent experiments. **: $p < 0.01$, *: $p < 0.05$. **e**, Representative bright-field imaging of the tumors. MCF7 cells stably expressing indicated shRNA were implanted into a subcutaneous site in the skin on one flank of athymic nude mice ($n=8$). On 4 weeks, mice receiving transplants of indicated cells were sacrificed. Scale bar, 1 cm.

f, Whisker plots showed the distribution of the tumor weights ($n=8$ tumors from 8 independent mice). Data were shown as mean $\pm$ s.d. and analyzed using Kruskal-Wallis test. **g**, WT and TG MEFs were serum-starved and treated with EGF for various times. Quantification of the phosphorylation levels of Akt, p38 and Erk protein relative to total Akt, p38 and Erk protein levels for Fig. 5i. Data are shown as mean $\pm$ s.d.. $n=3$ independent experiments (two-way ANOVA test). **h**, The mating strategy of OTUD3 TG mice crossing with MMTV-PyMT mice. Panels a,g are representative results of 3 or more independent experiments. 3×10^3 Statistics source data can be found in Supplementary Table 4. Uncropped images of blots are shown in Supplementary Fig. 9.



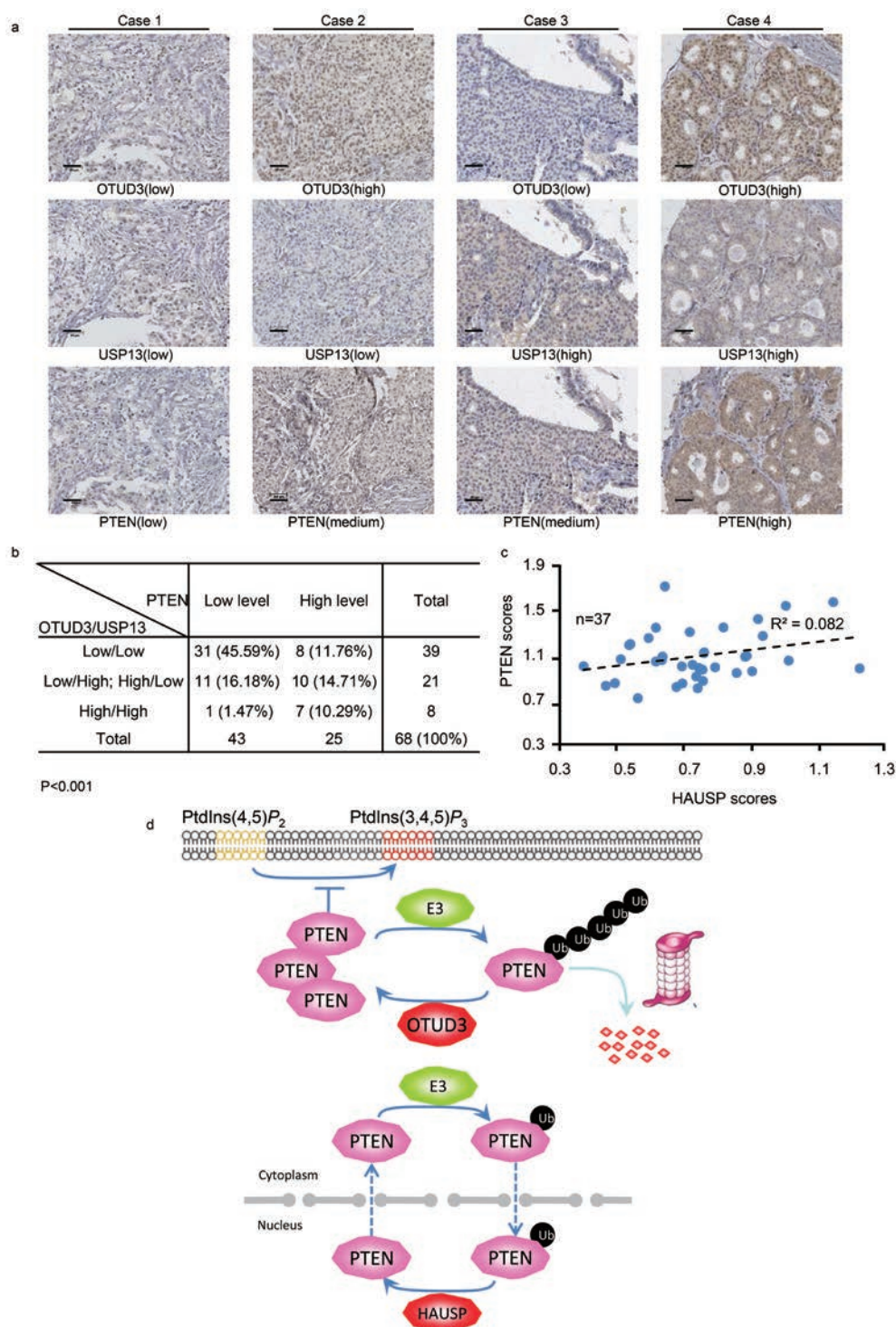
Supplementary Figure 6 OTUD3 E86K mutation in OTU domain leads to PTEN protein destabilization and increased tumorigenesis. **a**, Sequencing the region encoding OTUD3 OTU domain from 50 cases of Chinese breast cancer patients. Breast cancer tissues ($n=50$ independent samples) and the corresponding adjacent tissues ($n=50$ independent samples) were collected and subjected to RNA extraction and reverse transcription. The cDNAs were then sequenced. If the original sequencing results displayed heterozygous peaks, the corresponding cDNAs were ligated with pGEM-T vector (Promega)

and transformed into the competent bacteria. Then the colonies were sent out for sequencing. **b**, The mutation sites of OTUD3 in domain structure. Amino acid sequence alignment spanning OTUD3 R79 and E86 across species. **c**, Half-life of PTEN in the presence or absence of ectopic OTUD3 mutants. Quantification of the PTEN protein levels relative to β -actin for Fig. 6d (**: $p < 0.01$, two-way ANOVA test). Data are depicted as bar graphs with mean $\pm$ s.d.. $n=3$ independent experiments. Statistics source data can be found in Supplementary Table 4.



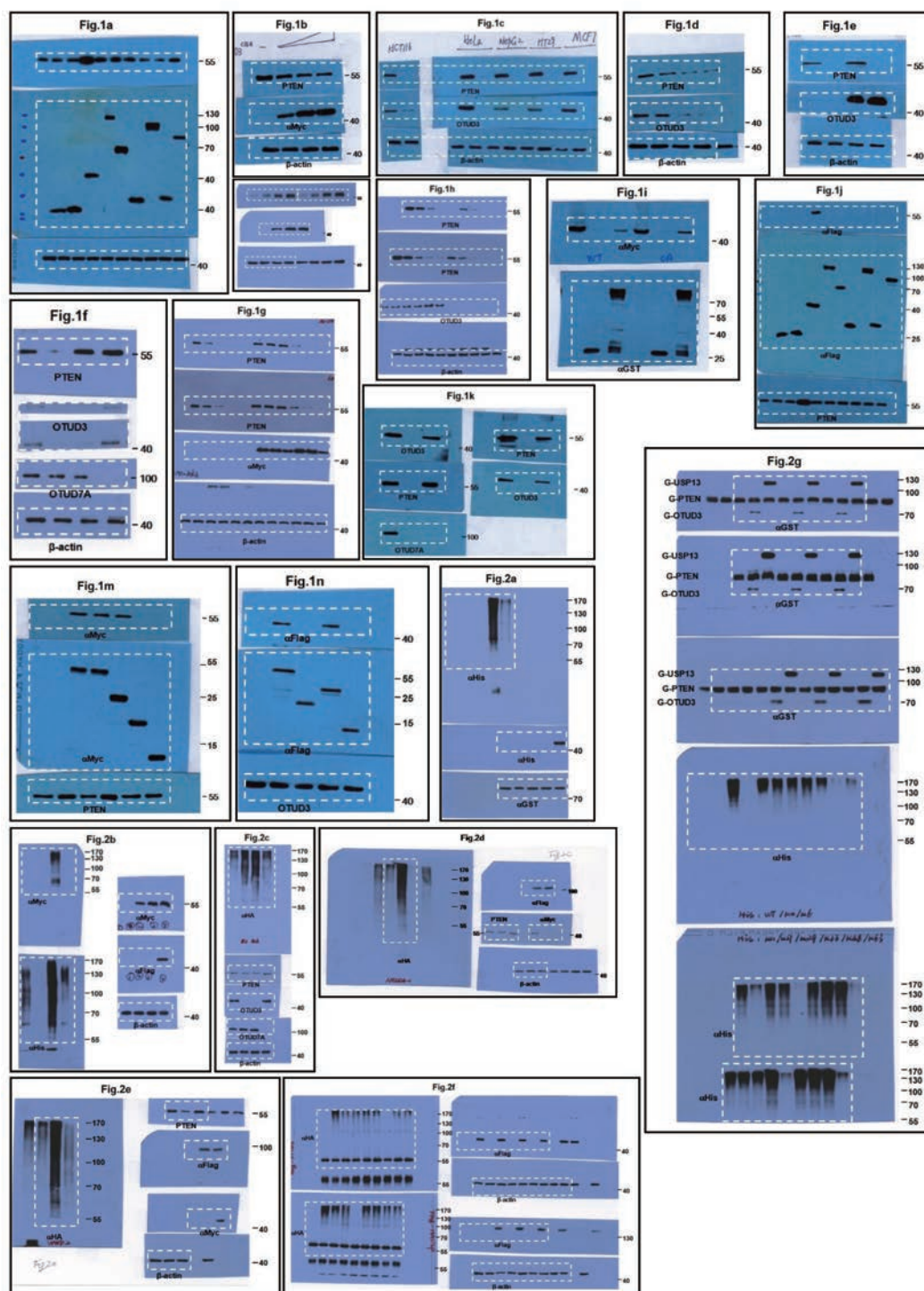
Supplementary Figure 7 OTUD3 expression is reduced in human breast cancer, which has no correlation with estrogen receptor, progesterone receptor or HER-2 expression. **a**, Relative OTUD3 mRNA level in breast cancer tissues and matched normal control from 30 subjects. Data were analyzed using Chi-square test. **b-d**, Whisker plots show the OTUD3 mRNA level in the cancer tissues classified into negative and positive groups based

on estrogen receptor (**b**), progesterone receptor (**c**) or HER-2 (**d**) expression from 50 subjects. Data were analyzed using Mann-Whitney test. **e**, Representative images from immunohistochemical staining of OTUD3, PTEN and p-Akt(S473) in three serial sections of the same tumor and matched adjacent tissue. The boxed areas in the left images were magnified on the right. **A**, adjacent tissue; **C**, carcinoma. Scale bar, 50 μm .

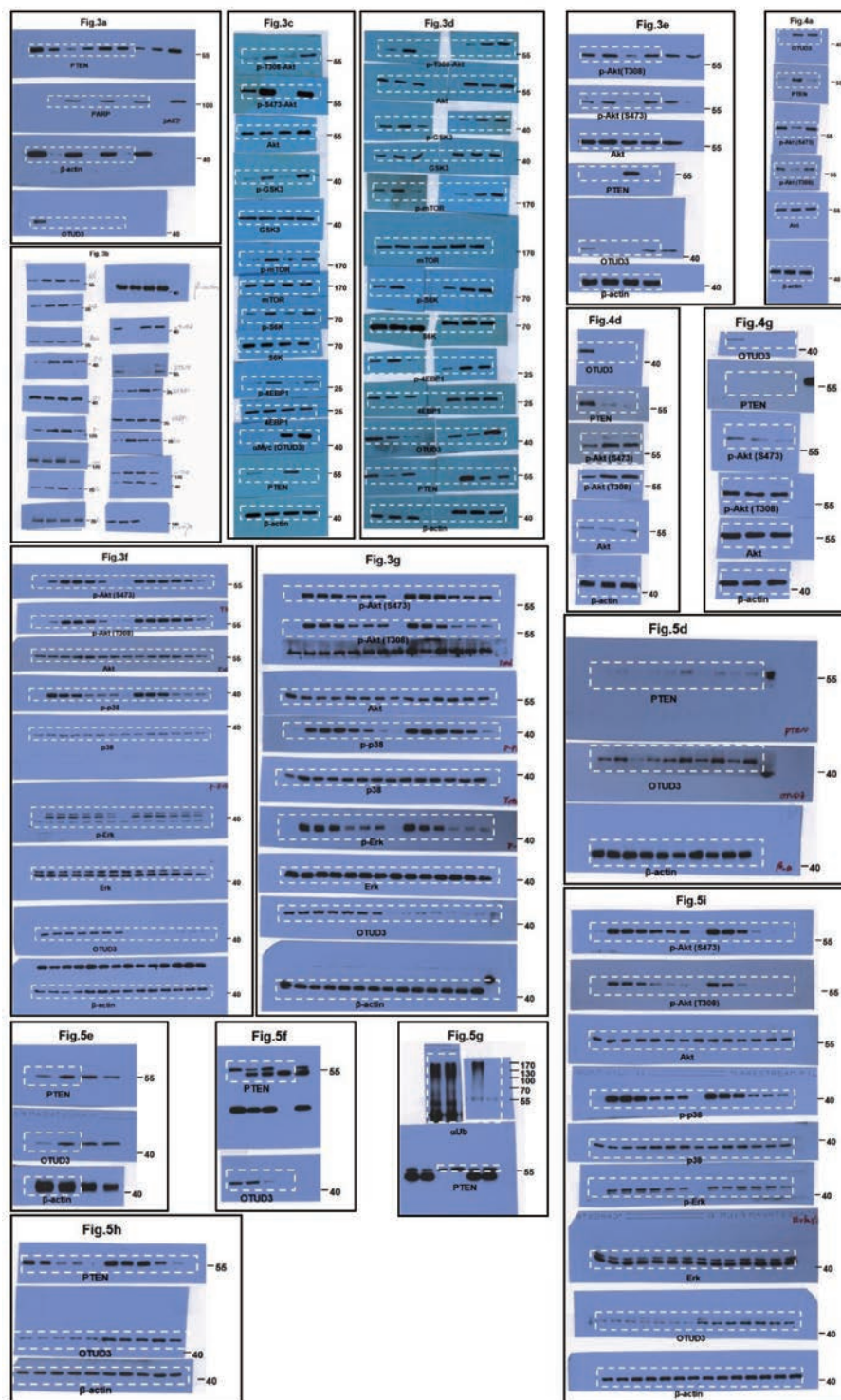


Supplementary Figure 8 OTUD3 and USP13 exhibit a synergy effect on PTEN expression in breast cancer samples. a,b, Potential synergy between OTUD3 and USP13 for PTEN protein levels. Serial sections of tissue arrays 68 patient breast specimens were subjected to immunohistochemistry with anti-OTUD3, anti-USP13 and anti-PTEN antibodies, respectively, and visualized by the DAB staining before imaging. Scale bar represents 50 μ m. Representative images were shown in (a), and the summary of the IHC results was listed in (b). $p < 0.001$, calculated by both Chi-square and χ^2 test. Scale bar, 50 μ m. c, Regression analysis comparing HAUSP and PTEN expression in breast cancer tissues. $n=37$. d, Proposed working model of OTUD3 on PTEN stability control.

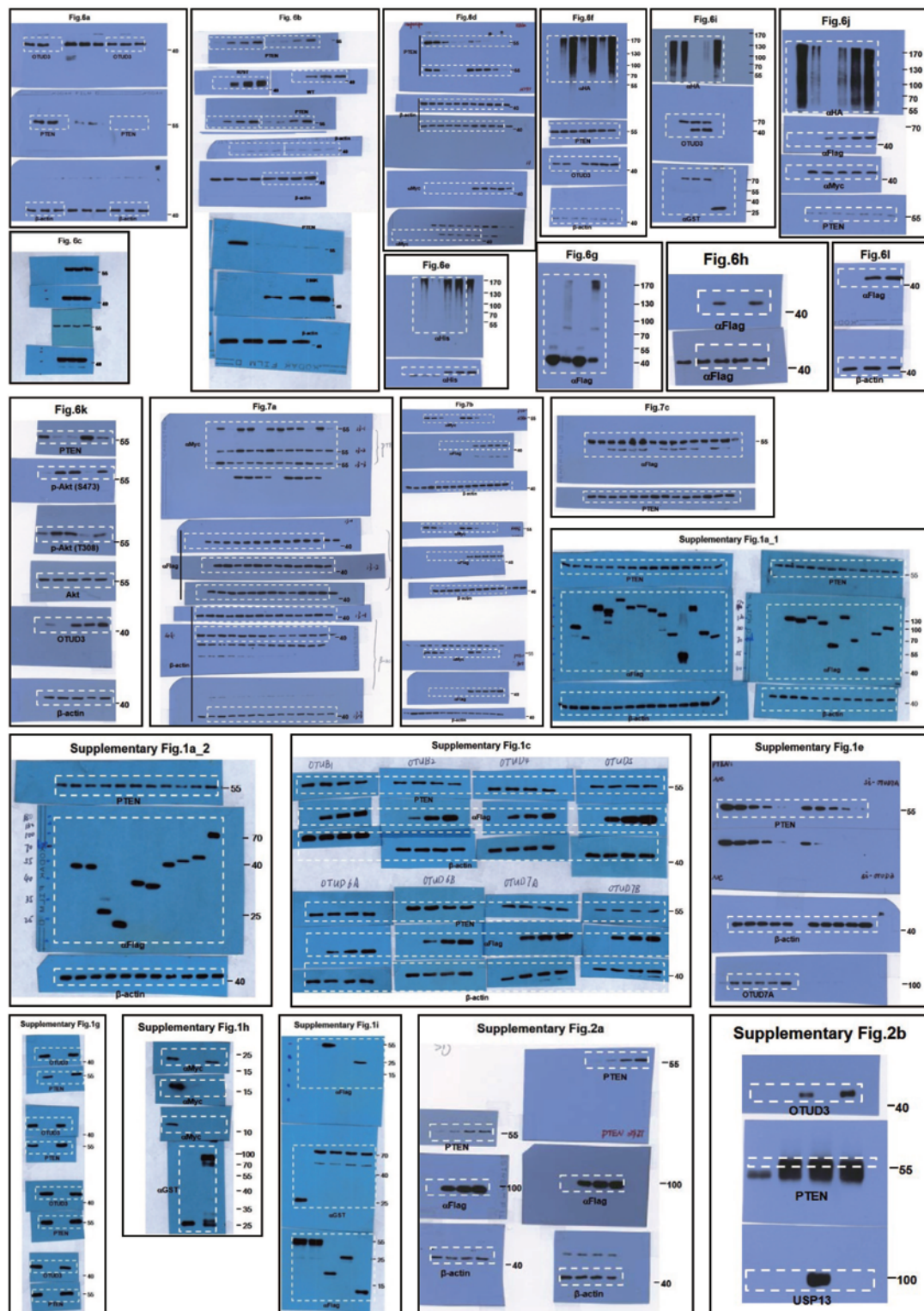
PTEN is a phosphatase that catalyzes the conversion of the lipid second messenger PtdIns(3,4,5)P₃ to PtdIns(4,5)P₂ and inhibits PI3K-Akt signaling. PTEN protein is relatively stable. This study identifies the deubiquitinase OTUD3 catalyzes the removal of poly-ubiquitin chain of PTEN in the cytoplasm. OTUD3 reverses the poly-ubiquitination of PTEN by E3 ubiquitin ligases (such as Nedd4-1, WWP2, XIAP and CHIP) and prevents PTEN degradation by 26S proteasome. On the other hand, a previous study (Ref. 22) identified that the deubiquitinase HAUSP is mainly localized in the nucleus and catalyzes the removal of mono-ubiquitination of PTEN. HAUSP regulates the PTEN cytoplasm-nucleus shuttling but has no effects on PTEN stability.



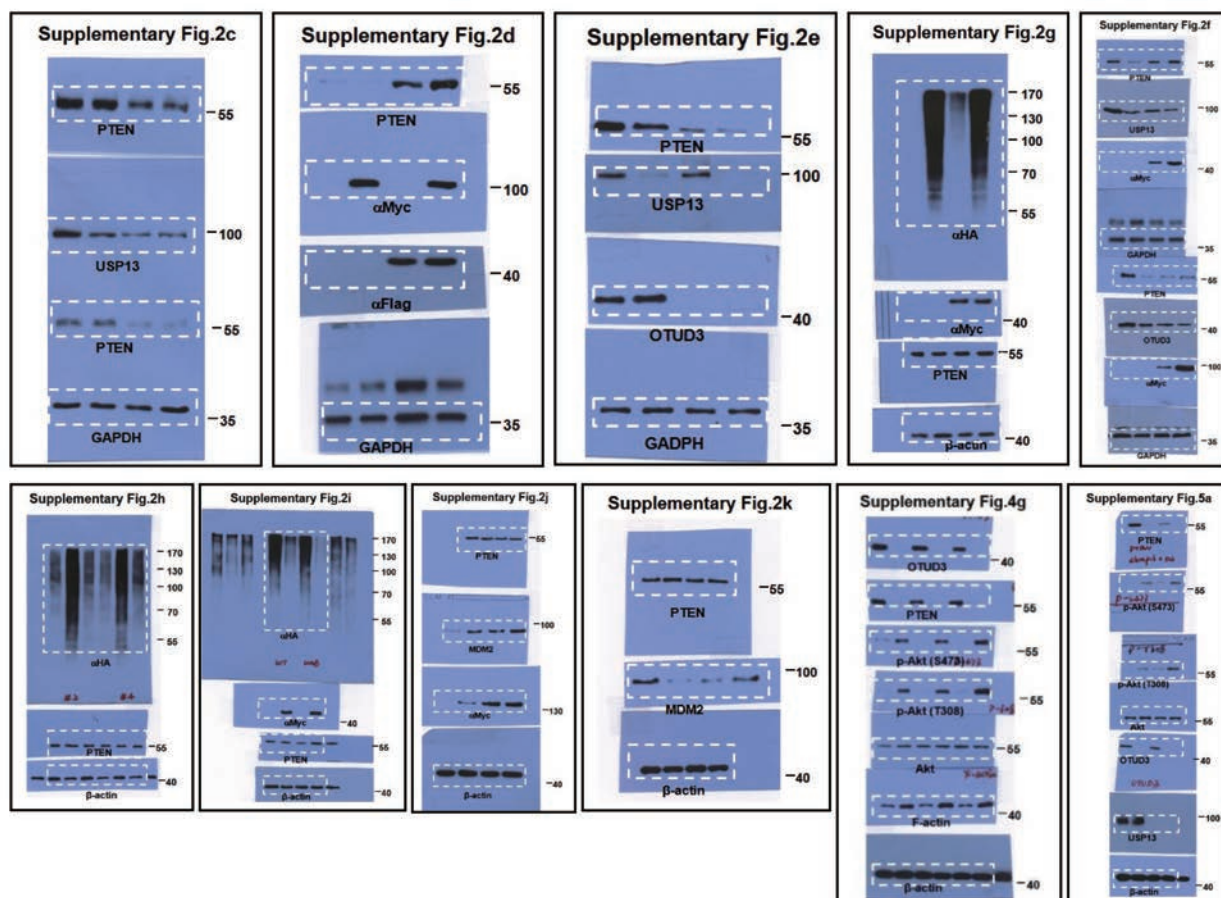
Supplementary Figure 9 Uncropped images of blots in Figures 1-7 and Supplementary Figures 1-5.



Supplementary Figure 9 continued



Supplementary Figure 9 continued



Supplementary Figure 9 continued

Table 1. OTUD3 mutations in human cancer. '+' shaded in light green represents the mutation can be queried in the databases.

	COSMIC	TCGA	CBIOPORTAL	CANCER
S89fs		+	+	Glioma
R90Q	+			Skin
K94R	+			Lung
R96K	+	+	+	Cutaneous Melanoma
R107L		+	+	Cutaneous Melanoma
E108V	+	+	+	Lung Adenocarcinoma
D109Y	+	+	+	Pancreas, Endometrial cancer
V114L	+			Skin
R144I	+	+	+	Endometrial cancer, Breast Invasive Carcinoma
A176T	+			Kidney
G180V	+	+	+	Kidney Renal Clear Cell Carcinoma
D184G			+	Hepatocellular Carcinom
D184N	+	+	+	Endometrial cancer
R187W	+	+	+	Stomach Adenocarcinoma, Lung Adenocarcinoma
D191E	+	+	+	Nasopharyngeal Carcinoma, Breast Invasive Carcinoma
L199P	+	+	+	Bladder Urothelial Carcinoma
D226V	+			Breast
D227N	+	+	+	Stomach Adenocarcinoma
N241N	+			Skin
R271W	+	+	+	Cutaneous Melanoma
R277I	+	+	+	NCI-60 Cell Lines, Stomach Adenocarcinoma
G302S			+	Pheochromocytoma and Paraganglioma
P327H	+	+	+	Stomach Adenocarcinoma
P327L	+	+	+	Esophageal Adenocarcinoma
V341I	+	+	+	Stomach Adenocarcinoma
T342fs		+	+	Cervical
Q350R			+	Hepatocellular Carcinom
Q358E	+			Thyroid
E359V	+			Thyroid
R361G	+			Thyroid
R363C	+	+	+	Colorectal Adenocarcinoma
E368G	+			Thyroid
S369R	+			Thyroid
R370S	+			Thyroid
N384S	+	+	+	Stomach Adenocarcinoma
607-2A>T	+			Kidney

Supplementary Table 1 OTUD3 mutations in human cancer. '+' shaded in light green represents the mutation can be queried in the databases.

Table 2. Primers for OTUD3 mutation selection in human breast cancer

Gene (H.sapiens)	Forward	Reverse
OTUD3-Exon 1	TTACCTTCCCAACGCTTGAGG	GCCGCCCCGAATGAATGGATG
OTUD3-Exon 2	GATCTGGCATGAAGCATCATTG	ATGCCAGTGGACTTAATTACCC
OTUD3-Exon 3	TGGAACTCAGTAAATACCAAAGC	CCCCTCATCTTCGGATATTCC
OTUD3-Exon 4	GTGGGTGTTTCTCTCCAGAAG	TCAGGCCCACAGATTCTCAATG
OTUD3-OTU (mRNA)	ATGTCCCGAAAGCAGGCGGCG	TTGATTCACTTGATGAAGCATCTG

Supplementary Table 2 Primers for OTUD3 mutation selection in human breast cancer.

Table 3. Primers for qPCR analysis

Gene (H.sapiens)	Forward	Reverse
OTUD3	TAAAGCAGCGGGAAGATTTTGA	TGCGATGTGTAACCTCCCTCAC
PTEN	TGGATTCTGACTTAGACTTGACCT	GGTGGGTATGGTCTTCAAAAGG
p21 ^{Cip1}	CGATGGAACTTCGACTTTGTCA	GCACAAGGGTACAAGACAGTG
p27 ^{Kip1}	AACGTGCGAGTGTCTAACGG	CCCTCTAGGGGTTTGTGATTCT
p15 ^{INK4b}	AGTCAACCGTTTCGGGAGGC	ACCACCAGCGTGTCCAGGAAG
VEGF	AGGGCAGAATCATCACGAAGT	AGGGTCTCGATTGGATGGCA
Bim	TAAGTTCTGAGTGTGACCGAGA	GCTCTGTCTGTAGGGAGGTAGG
p14ARF	GCCAGGGGCGCCCGCCGCTG	GGCCCGGTGCAGCACCACCA
p16	GCTGCCCAACGCACCGAATA	ACCACCAGCGTGTCCAGGAA
HAUSP	CCCTCCGTGTTTTGTGCGA	AGACCATGACGTGGAATCAGA
GAPDH	GGAGCGAGATCCCTCCAAAAT	GGCTGTTGTCATACTTCTCATGG
β-actin	CCTGGCACCCAGCACAAT	GCCGATCCACACGGAGTACTT

Supplementary Table 3 Primers for qPCR analysis.

Supplementary Table 4 The statistics source data for Figures and Supplementary Figures.